ALBANY
ASSA ABLOY

ACS 50 Control Unit



6410R0035

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These operating instructions are an integral part of the door. The original version of the operating instructions is in German.

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1 Introduction

These operating instructions are for the use of an ACS 50 control unit with a RapidRoll high-speed door. They constitute part of the product and describe how to use the product properly and safely throughout the product life cycle.

1.1 Applicable documents

These operating instructions are only valid in combination with the operating instructions for the RapidRoll high-speed door and the inspection book. Both of these documents are supplied with the product.

1.2 Declaration of conformity

The control unit is an integral part of the door. The declaration of conformity applies to the high-speed door as well as the control unit. It is included in the inspection book supplied with the door.

The control unit is designed in compliance with the requirements of the EN 13241-1 standard for door products and the subordinate standards mentioned in this standard.

Modifications that affect the technical data specified in the operating instructions or proper use, i.e. modifications that materially alter the machine, invalidate the manufacturer’s declaration of conformity.

2 Safety

2.1 General safety instructions

In addition to the general safety instructions in this section, always observe the safety instructions related to specific actions that are included in other sections. In particular, the general safety instructions in the operating instructions for the door are also applicable.

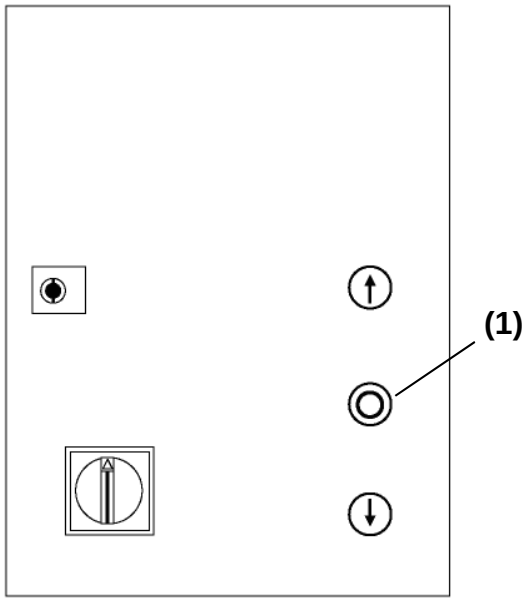
- All work on electrical systems and connections must be performed by qualified personnel with appropriate authorisation.
- Before performing any work on the control unit, cables, connectors or electrically live components, always switch off and lock the main switch or disconnect and secure the CEE plug.
- Do not operate the door unless the electrical supply to the door is equipped with a mains disconnect device (such as a suitable main switch or a CEE plug) that allows the electrical supply to the door to be disconnected. The mains disconnect device must be readily accessible and located between 0.6 and 1.7 metres above the entry level.
- The following conditions must be fulfilled to avoid endangering safe operation of the door:
  - The door is assembled in accordance with the installation instructions.
  - All wiring between the door, the control unit and the drive is connected properly.
  - All external impulse activators and safety devices are connected properly.
  - Unused inputs (for normally closed contacts) are bridged in accordance with the circuit diagram.

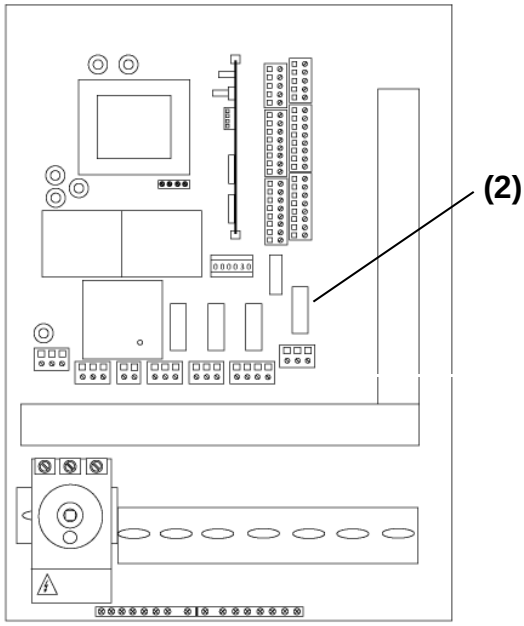
2.2 Safety devices

The closing sequence of a automatic door is monitored by various safety devices in order to avoid accidents. The safety devices and functions incorporated in the control unit are described below. The doors are also equipped with other safety devices, depending on the door model. See the 'Safety Systems' section of the RapidRoll user manual for information on the safety devices provided with the various models.

Depending on specific local conditions and especially when the door is operated by pedestrians, safety devices or impulse activators in addition to those specified in the standard may be necessary. In case of doubt, consult our sales engineers and obtain local advice.

2.2.1 Malfunction warning light and alarm relay





1	Malfunction warning light	2	Alarm relay

The red malfunction warning light **(1)** on the control box indicates a variety of malfunctions and operating states. In power-up mode and during self-testing, the lamp blinks for approximately three seconds. If a faults occurs during normal operation, it is indicated immediately and the alarm relay **(2)** drops out (see Section5.2.2).

2.2.2 Testing the safety devices

Before each door closing sequence, the microprocessor tests the light beams by briefly switching them off and on again. The microprocessor assesses the result of the complete test cycle. If the measured value differs from the expected value, a fault is present and the closing sequence is not started. Light beam testing is indicated by LED 24 going dark briefly and LEDs 27 and 28 on the control board blinking briefly (see Section 5.2.2).

In this state, the door can only be closed under manual control (hold to run mode).

WARNING

Risk of injury

All safety devices are disabled in hold to run mode. The door moves as long as the button on the control box is pressed and the door has not reached the upper or lower end position. The door does not switch off until it reaches the upper or lower end position.

⇨ Do not press the button if any person or object is located in the door hazard area.

2.2.3 Breakaway detector



For doors with manually repairable breakaway (RR300, RR300+ and RR392), the Breakaway contact is connected to the Stop input of the control unit. In this case breakaway must be manually reset before the door can be operated again.

Some door models can be fitted with breakaway capability on the bottom beam to avoid damage from vehicle impact. The door is stopped immediately if a vehicle hits the door and the breakaway detector is triggered. The malfunction warning light on the control box also lights up. In this state, the door can only be operated in hold to run mode.

Proceed as follows to restore normal operation:

1. Press and hold the Close button on the control box until the breakaway mechanism is reset.
2. Open the door completely and then close it.
3. Inspect the door carefully for any damage.

2.2.4 Run time monitoring

The control unit monitors the duration of the opening or closing sequence. If the duration of the sequence is longer than 20 seconds, the control unit responds as follows:

- The drive unit is stopped immediately to avoid damage to the drive and the door.
- The control unit is locked out to prevent it from processing any further impulses.
- The red malfunction warning light on the control box lights up.

Proceed as follows to restore normal operation:

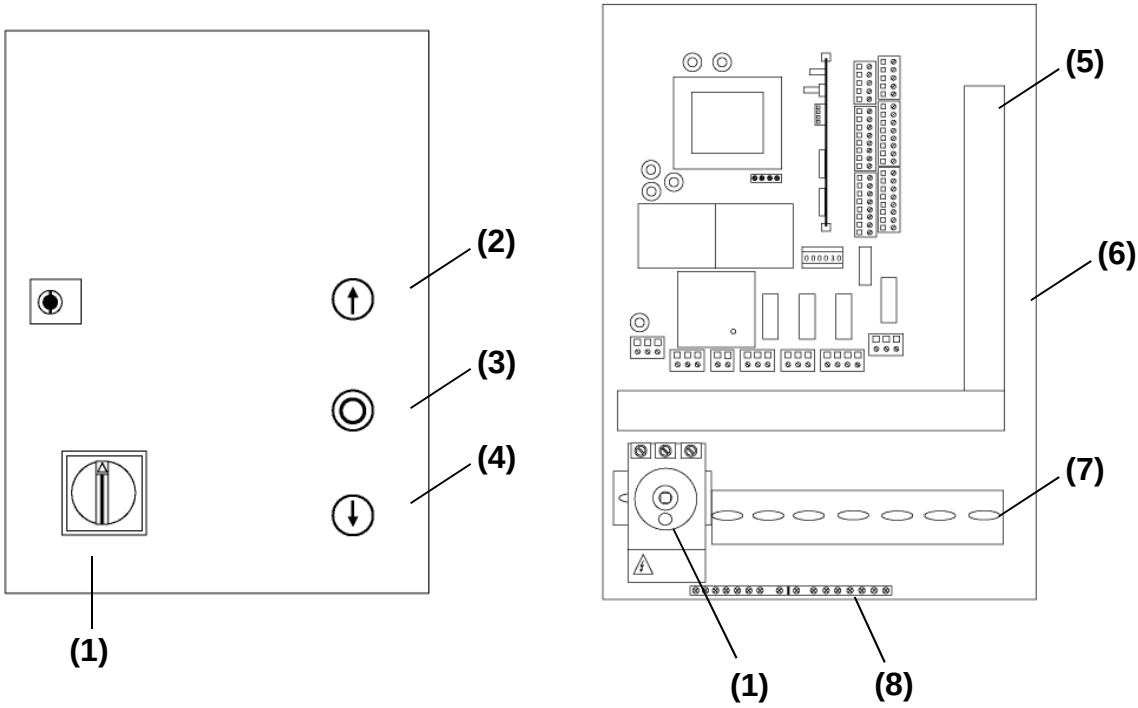
1. Correct the fault condition.
2. Reset the control unit by switching the main switch off and on again.

3 Product Description

The ACS 50 control unit is designed to be used with roller doors and folding doors. It is TÜV tested and conforms to the CE provisions. The control unit contains all the components necessary to control these doors with proper attention to all safety criteria.

3.1 Control unit layout

3.1.1 Control box



1 Main switch (emergency off)
2 Control button: Open
3 Control button: Stop
4 Control button: Close
5 Cable conduit
6 Mounting plate
7 Hat rail
8 PE rail

If the door cannot be seen from the control box installation location, an external control unit with a view of the door must be connected to the control unit in place of the three buttons on the control box. The external control unit consists of an enclosure with three buttons that assume the functions of the control elements on the control box.

3.1.1.1 Operational control elements of the control box

The control unit is switched on and off by the red main switch, which also serves as an emergency off switch. The main switch can be locked in the Off position by pressing the grey bar and fitting a padlock or similar device.

The three buttons **(2)**, **(3)** and **(4)** are used to operate the control unit in all operating modes and to reset it in the event of a fault **(4)**. The Stop button also has a lamp that provides a visual fault indication in case of a malfunction. This button always takes precedence over external impulse activators.

For doors equipped with induction loops acting as opening impulse activators, a centre-off rotary switch is fitted in place of the Open button. Rotating the switch to one side (momentary action) causes an opening impulse to be generated, while rotating it to the other side switches the door to the permanently open state.

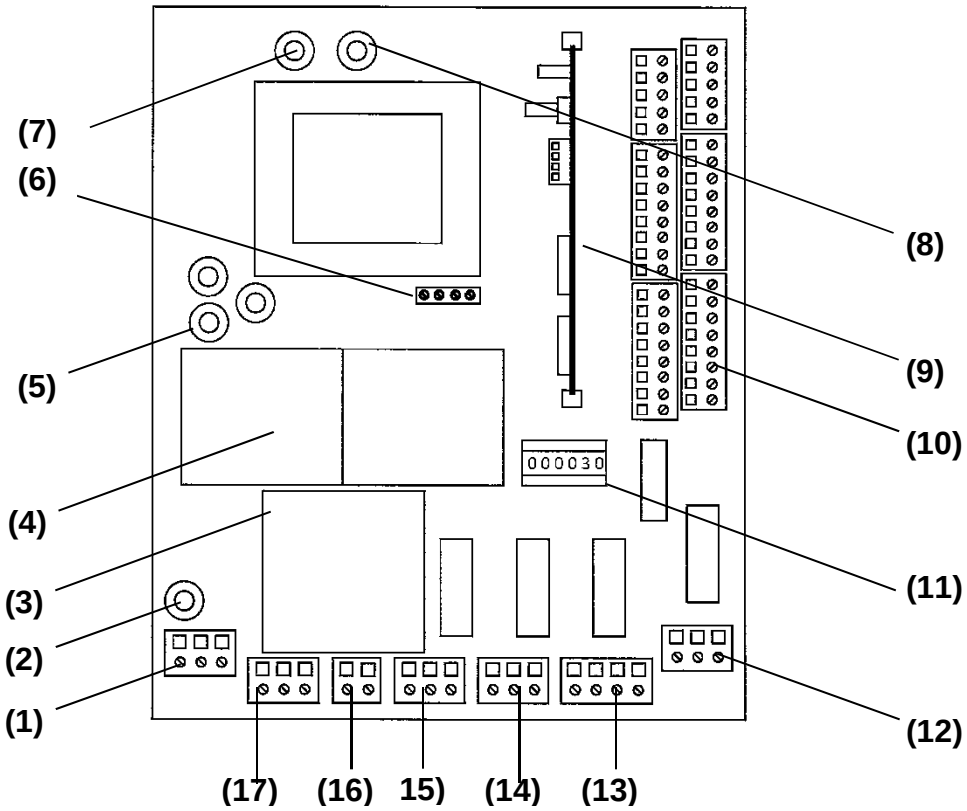
A rotary switch is also fitted when two interlocked control units are used. In this case, an opening impulse is also generated if the switch is rotated to one side. The switch latches when it is rotated to the other side. In this position, the associated door is unlocked and set to the permanently open state, while the other door can be used separately as an individual door (see Section 3.2.6.1 and Section 3.2.6.2**)**.

3.1.1.2 Control box components

There is a mounting plate **(6)** inside the control box, and the actual control components are located on a printed circuit board (motherboard and function card) fitted on this plate. A cable conduit **(5)** for the control wiring is located to the right of the circuit board. A screw-mounted hat rail **(7)** is located at the bottom of the mounting plate, and the main switch **(1)** is attached to the hat rail. Radio receivers, induction loop detectors and components for special functions can also be attached to the hat rail. A PE rail **(8)** is fitted below the hat rail. Cables are fed into the control box from below through cable glands.

3.1.2 Motherboard

The following modules are located on the motherboard:



1 Mains voltage terminals for external loads (protected by F4).
2 Mains voltage fuse for control voltage (F4, T1 A)
3 Plug-in brake rectifier
4 Motor contactors for driving the motor in the Open or Close direction
5 Mains inlet fuses (F1–F3, T6.3 A / 400 V & T10 A / 230 V)
6 Jumpers for mains voltage selection
7 Control voltage fuse 8 V AC (F5, T0.5 A)
8 Control voltage fuse 24 V DC (F6, T1A)
9 Socket for function card
10 Control unit terminals
11 Door cycle counter
12 Alarm relay terminals
13 Signal light terminals (traffic light / flashing light)
14 Terminals for door closed end position
16 Brake terminals
15 Terminals for door open end position
17 Motor terminals

3.1.2.1 Motherboard relay

There are four signal relays on the motherboard for signal outputs. The signal relays have potential-free changeover contacts (see the connection description in Section 6.1).

The signal light control relay has a normally open contact with dual terminals. The maximum rated contact load is 250 V AC / 8 A.

Relays	***Relay contact closure conditions***	
Door open end position	Door opened and upper end position reached	
Door closed end position	Door closed and lower end position reached	
Signal light control (traffic light and flashing light)	SW 1-4 ON	During door opening motion and during door closing motion with a pre-run period of 2 s.
	SW 1-4 OFF	During door closing motion with a pre-run period of 2 s
Fault / Operational	Control unit in good working order after power-up	

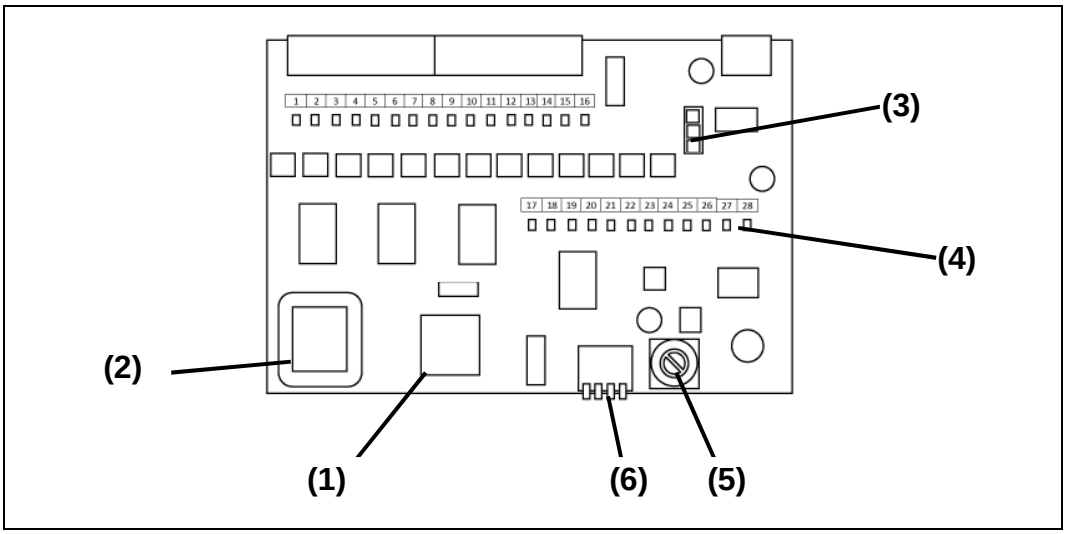
3.1.2.2 Brake rectifier

The control unit is designed for a brake voltage of 196 V. The brake rectifier must be chosen according to the applied mains voltage. The built-in LED can be used to distinguish the different rectifier types.

Supply voltage	***Brake voltage***	***Matching rectifier***
220/230 V	196 V	Bridge rectifier (with red LED)
380/400/415 V	196 V	Half-wave rectifier (with green LED)

A relay module is available for drives with integrated brake rectifiers. This relay module switches the external DC voltage from the rectifier built into the drive unit for the brake. The relay module does not have a LED.

3.1.2.3 Function card



1 Microcontroller
2 Program memory (EPROM)
3 Safety edge selection jumper
4 Indicators (LEDs)
5 Hold-open period selector switch
6 Operating mode programming switch

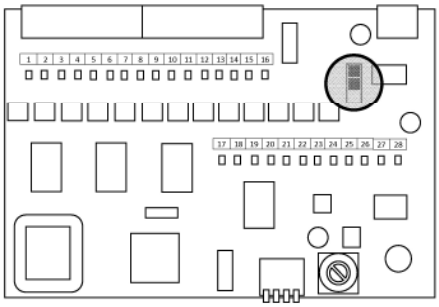
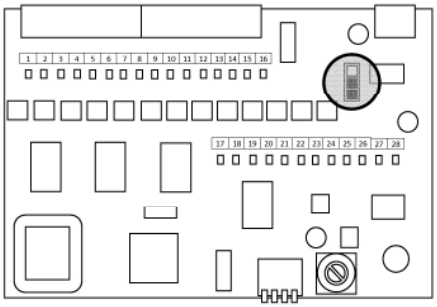
The safety edge signal is normally evaluated using an 8.2 kΩ termination resistor. This evaluation method is used with all newly manufactured doors.

As an alternative, the function card can be used with diode signal evaluation. With this option, the safety edge signal is evaluated using diode termination. Both cards have the same component structure and layout.

3.1.2.4 Closing edge protection

The control unit can be used with an electrical safety edge (8.2 kΩ resistor) or a pneumatic safety edge. This setting is configured by a jumper on the function card.

The closing edge input of the control unit fulfils the requirements of Category 2, Performance Level C of the EN ISO 13849-1:2008 standard, 'Safety of Machinery – Safety-related Parts of Control Systems'.

	
Electrical safety edge with 8.2 kΩ resistor	Pneumatic safety edge

3.1.2.5 Pre-running safety beam

The pre-running safety beam input fulfils the requirements of Category 2, Performance Level C of the EN ISO 13849-1:2008 standard, 'Safety of Machinery – Safety-related Parts of Control Systems'.

3.1.2.6 Emergency stop

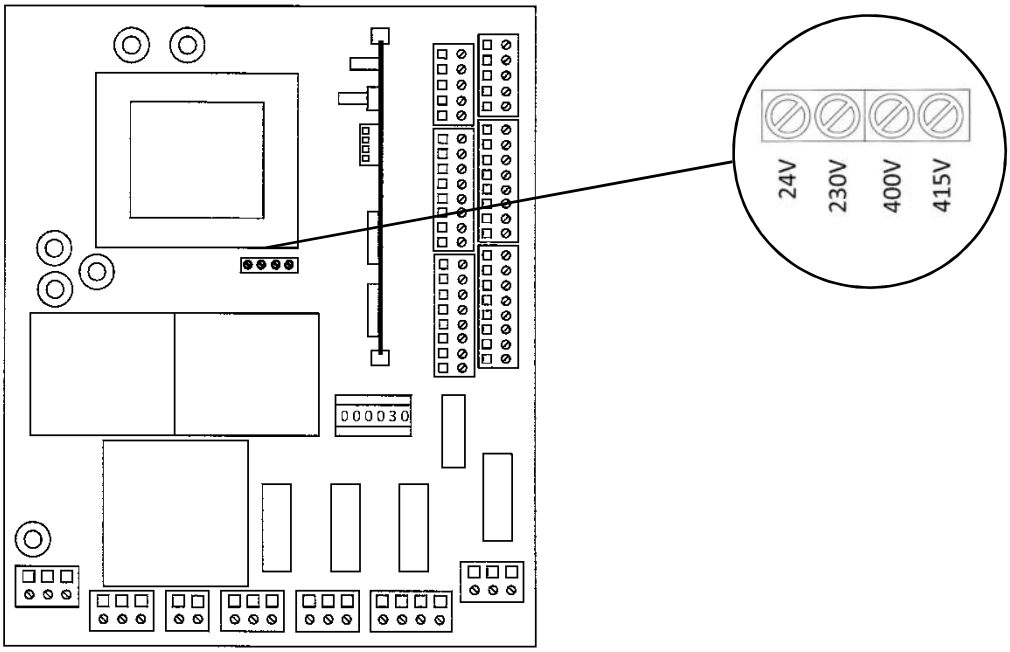
The ACS 50 control unit has provision for connecting two emergency stop control devices corresponding to Stop Category 0 (stopping by means of immediate interruption of the power supply to the motor).

The input for this fulfils the requirements of Category 1, Performance Level C of the EN ISO 13849-1:2008 standard, 'Safety of Machinery – Safety-related Parts of Control Systems'.

3.1.2.7 Jumpers for mains voltage selection

The mains voltage can be selected by jumpers on the motherboard according to the actual mains voltage.

The 24 V mains voltage selection is currently unused and has no effect.



3.1.2.8 Door cycle counter

A complete door cycle consists of an opening motion and a closing motion. The door cycle counter is fitted on the motherboard. The number of door cycles can be read from the counter.

3.1.2.9 Mains voltage terminals for external loads

The supply voltage depends on the applied mains voltage, since it is taken directly from an external line. With a 400 V mains voltage, the external line voltage is 230 V.

A supply voltage for external loads can be taken from the T3 and N terminals of the control unit. It is protected by fuse F4 (1AT).

3.2 Operating modes

The control unit has several operating modes, which can be selected by means of suitable configuration of the function card (see Section 4.3.1).

3.2.1 Open on impulse / close after time

After an Open impulse, the door travels until it reaches the door open end position. The closing sequence starts after the hold-open period expires. The door closes until it reaches the door closed limit switch.

If the door is closed, an Open impulse from the external impulse activator or the Open button on the control box causes the door to open immediately. A mushroom-head switch, pull switch, microwave or infrared motion detector, induction loop, wireless remote control, keypad, light beam, key switch, or other device with potential-free contacts can be used as an external impulse activator. If the door is open, the Open impulse restarts the hold-open period.

In response to a Stop command, either external or on the control box, the door stops immediately and remains where it is until motion in the desired direction is initiated by

another impulse activator. If the door is in the fully opened state, a Stop signal terminates automatic closing and the door remains open until the next impulse. If this is followed by a Close impulse, the closing sequence starts immediately. By contrast, an Open impulse restarts the hold-open period and consequently causes the door to close automatically after the period expires.

With door models equipped with a safety edge as a safety device on the bottom beam, DIP switch SW 1-3 must be set to ON. With this setting, the door will stop immediately if the safety edge is actuated and then open until it reaches the door open end position. The door is locked in this position and remains open until the door closing sequence is initiated at the control box.

DIP switch S! 1-3 is set to OFF for doors with pre-running light beams. If the pre-running light beam is interrupted, the door also reverses direction, but it closes again after the hold-open period expires.

If the signal from the stationary light beam or another safety device is interrupted, the door responds in the same way as with the pre-running light beam.

3.2.2 Toggle circuit

The toggle circuit allows the door to be opened and closed using a single impulse activator. As long as the door is not fully open, every incoming impulse is interpreted as an Open impulse. An impulse received when the door is open starts a closing sequence. For safety reasons, only manual impulse activators (pull switches or buttons) should be used with this function.

The safety devices work the same way as in the 'open on impulse / close after time' operating mode. The close after passage function cannot be used in this operating mode. It is possible to switch between the signalling options.

3.2.3 Close after passage function

If the close after passage function is enabled, the door is opened by an Open impulse activator. The closing sequence is initiated by the interruption of the stationary safety light beam. The door closes with a 2-second delay after the stationary light beam is first interrupted and then unobstructed.

The close after passage function can also be combined with the close after time function. With this arrangement, the door closes after the hold-open time if the stationary light beam is actuated.

In case of doors with limit switches and a stationary light beam at the closing level, the limit switch for disabling the stationary light beam must be configured as follows: enable the close after passage function and open the door. If the car-park period is activated automatically after this, the switch-off point of the stationary light beam must shifted upward slightly (see the installation instructions).

3.2.4 Hold to run mode

The door travels as long as the button on the control box is pressed and it has not reached the upper or lower end position. The door switches off automatically when it reaches the upper or lower end position. All safety devices are disabled in hold to run mode.

3.2.5 Interlocking

If two or more doors are interlocked, the control units are wired such that only one door can be opened at the same time. If one door is open or opening, all other doors are locked electrically and cannot be opened.

3.2.6 Options

3.2.6.1 Permanent opening

The Open button can optionally be replaced by a centre-off rotary switch for permanent opening. This switch is momentary-acting in one direction (Open impulse) and latching in the other direction (permanent opening). The switch is changed to the null position to terminate permanent opening.

3.2.6.2 Permanent opening switch with an induction loop

For doors with induction loops, a permanent opening switch is used to avoid the need to switch off the control unit for permanent door opening. Otherwise a new loop calibration would occur after the control unit is switched on. The objects and vehicles on the induction loop would be included in the calibration and consequently no longer detected.

3.2.6.3 Interlock disabling

With interlocked doors, the permanent opening switch can also be used to manually disable interlocking. The switch must be rotated to the latched position to disable interlocking. In this position, the other door acts as an individual door. The interlock function can be enabled again by rotating the switch back to the null position. The momentary-action position of the permanent opening switch can be used to open the door.

4 Setup and Initial Use

4.1 Supply circuit requirements

For proper operation of the control unit, the supply circuit must fulfil the following requirements:

- The mains supply cable must have a cross-sectional area of at least 5 x 1.5 mm^2. Conductors with cross-sectional areas up to 2.5 mm^2 can be connected.
- The back-up protective device must be selected according to the conductor cross-sectional area and the selectivity of the fuses in the control unit (T6.3 A / 400 V or T10 / 230 V).
- The 400 V / 50 Hz supply circuit from a 3/(N)/PE mains network must be arranged with a clockwise phase sequence. The control unit can optionally be connected to a 230 V, 400 V or 415 V supply voltage with clockwise phase sequence (see the circuit diagram for special models).
- The door must be connected to the potential equalisation rail.

4.2 Installation

	⚠	***CAUTION***

Wiring damage

Improper installation can cause damage to the wiring.

⇨ Ensure that the wiring is routed correctly during control box installation.

1. Install the control box according to local conditions:
   - o Attached to masonry using four M8x60 hexagon-head screws and 10-mm plugs
   - o Fitted on a steel substructure and secured with M8x30 screws
2. A order-specific circuit diagram is included with each door. Wire the control unit according to the included circuit diagram.
3. Configure the appropriate settings and perform the appropriate tests for normal door operation (see Section 4.3).

4.3 Settings and functional tests

1. Switch on the main switch.
2. Configure the limit switches for the door open and door closed positions according to the corresponding door positions by driving the door toward each position and adjusting the limit switch.
3. Test all of the safety devices by triggering them.
4. Test the impulse activators for proper operation.
5. Test the door for proper operation by means of several test runs.

The door and the control unit are now ready for use.

4.3.1 Setting the operating mode

DIP switches SW1-1 to SW1-4 set the operating mode of the control unit. The corresponding switch positions are shown in the following table.

SW1-1	SW1-2	SW2	***Terminals*** 3-4; 3A-4A	***Terminals*** 5-6; 5A-6A
ON	ON	0	Toggle circuit	Toggle circuit
ON	ON	1-F	Close after passage & Close after time	Toggle circuit
ON	OFF	0	Open	Toggle circuit
ON	OFF	1-F	Open & Close after time	Toggle circuit
OFF	ON	0	Close after passage function	Close
OFF	ON	1-F	Open & Close after time	Open & Close after time
OFF	OFF	0	Open	Close
OFF	OFF	1-F	Open & Close after time	Close

4.3.2 Setting the hold-open period

The hold-open period for automatic closing after a time delay is set using rotary hexadecimal switch SW2. See the following table for specific time settings.

SW2	***Automatic closing after the following hold-open period:***
0	Automatic closing disabled; closing sequence initiated by a Close impulse
1	1 s
2	2 s
3	3 s
4	4 s
5	6 s
6	7 s
7	10 s
8	13 s
9	16 s
A	19 s
B	22 s
C	25 s
D	30 s
E	45 s
F	60 s

5 Operation

5.1 Safety instructions

Observe the safety instructions in Section 2 and the safety instructions in the user guide for the door.

5.2 Operation

5.2.1 Program code and memory test

The control unit tests the stored program code and memory each time it is switched on. Control impulses are ignored during the test phase. If a fault is detected, the red lamp on the control box lights up, the alarm relay remains in the disengaged state, and the control unit is locked. If no fault is detected, the red lamp blinks for approximately 3 seconds to indicate that the unit is in power-up mode. The impulse must be interrupted to unlock the control unit. The desired function can only be triggered by the rising edge of the next impulse. The power-up mode ends after approximately 3 seconds, after which the control unit is ready for normal operation.

SW1-3: Locking



This setting must be set to ON (locking enabled) for a door with a safety edge and to OFF (locking disabled) for a door with a pre-running light beam.

This switch determines whether the door remains locked in the top position after the safety edge is triggered or starts a closing sequence after the hold-open period expires.

Signal light control:

SW1-4: ON = signal light active during opening and closing travel

The output relay for signal lamp control engages when an Open impulse is generated while the door is closed. It remains engaged until the door reaches the door open end position. After the configured hold-open period expires, the pre-run period for signal light control starts and the relay engages again. The duration of the pre-run period is 2 seconds, after which the door closing sequence begins. The signal light relay remains engaged until the door is fully closed.

SW1-4:OFF = signal light active only during closing travel

With this option, the relay closes when the hold-open period expires and the pre-run period starts. The closing sequence starts after the 2-second pre-run period expires. The relay remains engaged until the door reaches the door closed end position.

5.2.2 Operating mode and malfunction indication

The operating state of the door is indicated by 28 LEDs of different colours fitted on the function card.

- The green LEDs indicate that the operating state is fault-free and the connected operating equipment is in good working order.
- The yellow LEDs indicate motions or actions performed by the control unit or signalled externally to the control unit.
- The red LEDs indicate a fault or malfunction.

The relationship between the LEDs and the functions is shown in the following table, and the arrangement of the LEDs on the function card is described in Section 3.1.2.3.

LED	Function	Colour	Description
1	According to set operating mode	yellow	Lights up when an external impulse at terminal 4-4A is present
2	Stop button	green	Off when the Stop button is pressed
3	According to set operating mode	yellow	Lights up when an external impulse at terminal 6-6A is present
4	Upper limit switch	yellow	Lights up when the door open limit switch is actuated
5	Lower limit switch	yellow	Lights up when the door closed limit switch is actuated
6	Safety edge	red	Lights up when the safety edge is actuated
7	Light beam 1	green	Off if the safety device has triggered
8	Temperature switch	green	Off if the temperature switch has been actuated
9	Emergency manual lever	green	Off when the emergency manual lever is actuated
10	Open button on the control box	yellow	Lights up when the Open button the control box is pressed
11	Breakaway detector	green	Off when the breakaway detector is actuated
12	Spare input	red	Remains off
13	Close relay test	yellow	Lights up during closing travel
14	Close button on control box	yellow	Lights up when the Close button on the control box is pressed
15	Light beam 2	green	Off if the safety device has triggered
16	Pneumatic safety edge	yellow	Off when the safety edge is actuated
17	Opening contactor	yellow	Lights up while the door is opening
18	Closing contactor	yellow	Lights up while the door is closing
19	Brake	yellow	Lights up when the brake is actuated (electrically engaged)
20	Malfunction indication	red	Lights up when a malfunction is present
21	Traffic light control	yellow	Lights up when the traffic light is enabled
22	Alarm signal	green	Lights up when the control unit is operational
23	Spare output	yellow	Remains off
24	Safety devices test	green	Goes dark briefly while the safety devices are being tested; otherwise always on
25	Light beam fault signal	red	Lights up if the light beam is faulty
26	Safety edge fault signal	red	Lights up if the safety edge is faulty
27	Safety edge test	yellow	Lights up briefly during the safety edge self-test
28	Safety edge test	yellow	Lights up briefly during the safety edge self-test

Contact the supplier or manufacturer for fault correction and repairs.

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Certain combinations of the LEDs, control unit states and the malfunction warning light indicate malfunctions as follows:

Malfunction warning light	***LED & control unit state***	***Cause of fault***
Off		No supply voltage present at control unit
Blinking		Power-up mode (for three seconds after switch-on)
	LD9=off	Emergency manual handle actuated
On		Persistent impulse applied to the control unit during power-up mode
	LD6 on	Safety edge fault
	LD25 on	Short circuit in safety beam circuit
	LD7 off	Safety beam interrupted
	LD7 off & control unit locked out	temperature switch triggered
	LD26 on & SW1-3 on	Safety edge triggered
	Control unit locked out	Maximum running time exceeded
	Control unit locked out	EPROM test or RAM test fault
	LED27 or LED28 blinking	Blinking LED indicates a self-test fault or a safety edge fault

6 Appendix

6.1 Motherboard signal assignments

The signal assignments of the terminals on the motherboard are listed in the following table.

Terminal	***Assignment***
T1, T2, T3	Input voltage
T3, N	Supply voltage for external modules
U, V, W	Motor connection
70, 71	Brake
1, 2, 1A, 2A	Stop input
3, 4, 3A, 4A	External impulse inputs
5, 6, 5A, 6A	External impulse inputs
7, 8	Safety edge
9	+24 V DC to stop circuit
10	Door open limit switch
12	+24 V DC to stop circuit
13	Door closed limit switched
22, 22A	+24 V DC
20, 20A	0 V DC
21, 21A	+24 V DC with testing
23A	Safety device 1
23	Safety device 2
33, 33A	0 DC
34, 34A	+24 V DC
35, 36	Temperature switch
37, 38	Emergency manual lever
45, 46	Breakaway detector (NC contact)
53, 54	Breakaway detector (NO contact)
55, 56	Spare output
63, 64	24 V AC

Potential-free signals can be taken from the following terminals:

Terminal	***Assignment***	***Terminal***	***Assignment***
74, 75, 76	Door open	74	COM
		75	NO contact
		76	NC contact
77, 78, 79	Door closed	77	COM
		78	NO contact
		79	NC contact
80, 80A, 81, 81A	Signal light control	80, 80A	COM
		81, 81A	NO contact
82, 83, 84	Malfunction	82	COM
		83	NO contact
		84	NC contact

6.2 Technical Data

Dimensions of housing WxHxD	300 x 400 x 210 mm (Steel powder coated) 380 x 380 x 210 mm (Stainless steel)
Main voltage	3~ PE; 50-60 Hz, 230V, 380V, 400V, 415V
Max. output power	3 kVA
Operating temperature range	0℃ - +40℃
Protection class	IP 55
External fuse by customer	10A (220 V / 230 V: 16A)
Control voltage / external supply	24 VDC / max. 700 mA
Brake voltage	196 VDC
Weight	ca. 12 Kg
Conformity Europe	2006/42/EC, 2004/108/EC, 2006/95/EC EN13241-1: 2004, EN ISO 13849-1:2008 EN 12978:2009, EN 12453:2000 EN 60335-1:2002, TÜV Nord certification

6.3 Control unit ID



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